

TUTORIAL SHEET-03

Course: B.Tech (CSE)

Section: D/J/P

Subject: Computer Networks

Subject Code: BCSC0008

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Session: 2024-25

1. Assume that a **12-bit** Hamming codeword consisting of **8-bit data and 4 check bits** is **$d_8d_7d_6d_5c_8d_4d_3d_2c_4d_1c_2c_1$** , where the data bits and the check bits are given in the following tables:

Data bits								Check bits			
d_8	d_7	d_6	d_5	d_4	d_3	d_2	d_1	c_8	c_4	c_2	c_1
1	1	0	x	0	1	0	1	y	0	1	0

Which one of the following choices gives the correct values of x and y ?

- A. x is 0 and y is 0
 - B. x is 0 and y is 1
 - C. x is 1 and y is 0
 - D. x is 1 and y is 1
2. Sixteen-bit messages are transmitted using a Hamming code. How many check bits are needed to ensure that the receiver can detect and correct single-bit errors? Show the bit pattern transmitted for the message 1101001100110101. Assume that even parity is used in the Hamming code. (Assume position counting starts from left to right)
3. An error correcting code has the following code words: 00000000, 00001111, 01010101, 10101010, 11110000. What is the maximum number of bit errors that can be corrected?
4. A sender needs to send the four data items 0x3456, 0xABCC, 0x02BC, and 0xEEEE. Answer the following:
- a. Find the checksum at the sender site.
 - b. Find the checksum at the receiver site if there is no error.
 - c. Find the checksum at the receiver site if the second data item is changed to 0xabce.
 - d. Find the checksum at the receiver site if the second data item is changed to 0xABCE.
5. Consider a selective repeat sliding window protocol that uses a frame size of 1 KB to send data on a 1.5 Mbps link with a one-way latency of 50 msec. To achieve a link

utilization of 60%, the minimum number of bits required to represent the sequence number field is _____.

6. Station A uses 32 byte packets to transmit messages to Station B using a sliding window protocol. The round trip delay between A and B is 80 milliseconds and the bottleneck bandwidth on the path between A and B is 128 kbps. What is the optimal window size that A should use?
7. Suppose that the stop-and-wait protocol is used on a link with a bit rate of 64 kilobits per second and 20 milliseconds propagation delay. Assume that the transmission time for the acknowledgment and the processing time at nodes are negligible. Then the minimum frame size in bytes to achieve a link utilization of at least 50% is _____.
8. What is the total delay (latency) for a frame of size 5 million bits that is being sent on a link with 10 routers each having a queuing time of $2\mu\text{s}$ and a processing time of $1\mu\text{s}$. The length of the link is 2000 Km. The speed of light inside the link is 2×10^8 m/s. The link has a bandwidth of 5 Mbps. Which component of the total delay is dominant? Which one does dominant the transmission time?
9. Suppose a 128kbps point to point link is setup between earth and Rover on Mars. The distance from earth to Mars is approximately 55Gm data travel over the link at the speed of light. What is the minimum RTT for the link?
10. We have a pure ALOHA network with 100 stations. If $T_{fr}=1\mu\text{s}$, what is the number of frames each station can send to achieve the maximum efficiency. Write vulnerable time and throughput of pure and slotted aloha.
11. How pure ALOHA differ from slotted ALOHA. Discuss the vulnerable time of Pure ALOHA and slotted ALOHA. One hundred stations on a pure ALOHA network share a 1 Mbps channel. If frames are 1000 bits long, find the throughput if each station is sending 500 frames per second. How many frames will survive?
12. A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces
 - i. 1000 frames per second?
 - ii. 500 frames per second?
 - iii. 250 frames per second?
13. A network has a data transmission bandwidth of 20×10^6 bits per second. It uses CSMA/CD in the MAC layer. The maximum signal propagation time from one node to another node is 40 microseconds. The minimum size of a frame in the network is _____ bytes.
14. Consider a CSMA/CD network that transmits data at a rate of 100Mbps (10^8 bits per second) over a 1km (kilometre) cable with no repeaters. If the minimum frame size required for this network is 1250bytes, what is the signal speed (km/sec) in the cable?